

Lab 12: Flask Introduction

1. Flask Overview

Flask is a lightweight web framework written in Python. It allows you to build web applications quickly with minimal setup. Flask follows the MVC (Model-View-Controller) pattern and is great for both beginners and professionals due to its simplicity and flexibility.

2. Basic Structure of a Flask Application

A typical Flask project is structured as follows:

- **app.py**: The main Python file where the Flask application is defined and run.
- **templates/**: Contains HTML files (Jinja2 templating).
 - Example: index.html — the main web page.
- **static/**: Contains static files like CSS, JavaScript, and images.
 - Example: style.css, script.js

3. Setting Up Flask

To run a Flask application:

Step 1: Install Flask

Open your terminal or command prompt and run:

```
pip install flask
```

Step 3: Run the Flask App

Navigate to the project folder where app.py is located, and run the following:

```
python app.py
```

Step 4: Access the App

Once the app is running, open your browser and go to:

```
http://127.0.0.1:5000
```

Example: Flask Based Tic Tac Toe Game

This project creates a 2-player Tic Tac Toe game using Flask, with a glowing aesthetic grid made from clickable <div> elements. It shows the winner at the end and includes a "Reset Game" button.

Folder Structure:

```
project/  
|  
├── app.py  
├── static/  
|   ├── style.css  
|   └── script.js  
└── templates/  
    └── index.html
```

app.py

```
from flask import Flask, render_template  
  
app = Flask(__name__)  
  
@app.route('/')  
def index():  
    return render_template('index.html')  
  
if __name__ == '__main__':  
    app.run(debug=True)
```

templates/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Tic Tac Toe</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
  <h1>Tic Tac Toe</h1>
  <div id="gameBoard">
    {% for i in range(9) %}
      <div class="cell" onclick="makeMove(this)"></div>
    {% endfor %}
  </div>
  <div id="status"></div>
  <button onclick="resetGame()">Reset Game</button>

  <script src="{{ url_for('static', filename='script.js') }}"></script>
</body>
</html>
```

static/style.css

```
body {  
    text-align: center;  
    font-family: Arial, sans-serif;  
    background: #0f0f0f;  
    color: #fff;  
}  
  
h1 {  
    margin-top: 20px;  
    color: #00ffcc;  
    text-shadow: 0 0 10px #00ffcc;  
}  
  
#gameBoard {  
    display: grid;  
    grid-template-columns: repeat(3, 100px);  
    grid-gap: 10px;  
    justify-content: center;  
    margin: 30px auto;  
}  
  
.cell {  
    width: 100px;  
    height: 100px;  
    background: #222;  
    border: 2px solid #00ffcc;  
    display: flex;  
    align-items: center;  
    justify-content: center;
```

```
font-size: 36px;  
cursor: pointer;  
color: #00ffcc;  
box-shadow: 0 0 10px #00ffcc;  
transition: 0.3s;  
}
```

```
.cell:hover {  
    background: #333;  
}
```

```
#status {  
    font-size: 24px;  
    margin: 20px;  
    color: #ff66cc;  
}
```

```
button {  
    padding: 10px 20px;  
    font-size: 16px;  
    background: #ff66cc;  
    border: none;  
    border-radius: 5px;  
    color: #fff;  
    cursor: pointer;  
    box-shadow: 0 0 10px #ff66cc;  
}
```

static/script.js

```
let currentPlayer = "X";

let gameActive = true;

let board = [ "", "", "", "", "", "", "", "", "" ];

function makeMove(cell) {
    const index = Array.from(cell.parentNode.children).indexOf(cell);
    if (!gameActive || board[index]) return;

    board[index] = currentPlayer;
    cell.textContent = currentPlayer;

    if (checkWinner()) {
        document.getElementById("status").textContent = `${currentPlayer} Wins!`;
        gameActive = false;
    } else if (!board.includes("")) {
        document.getElementById("status").textContent = "It's a Draw!";
        gameActive = false;
    } else {
        currentPlayer = currentPlayer === "X" ? "O" : "X";
    }
}

function checkWinner() {
    const winPatterns = [
        [0,1,2], [3,4,5], [6,7,8],
        [0,3,6], [1,4,7], [2,5,8],
        [0,4,8], [2,4,6]
    ];
```

```
return winPatterns.some(pattern =>
  pattern.every(index => board[index] === currentPlayer)
);
}

function resetGame() {
  board = [ "", "", "", "", "", "", "", "", "" ];
  currentPlayer = "X";
  gameActive = true;
  document.querySelectorAll(".cell").forEach(cell => cell.textContent = "");
  document.getElementById("status").textContent = "";
}
```

Lab 12 Task

1. (Continue from Lab 11):

- Develop Model's Application Phase in Flask
- Use ChatGPT for the frontend code